0.0.1 $\langle \Delta H_{SD} \rangle$ Derivation

The equation $H = -q(1-q)$ defines a constant of motion for the SD part of the game, where q is the fraction of players in the 4th strategy. Using the transition probabilities of the different process we can derive an expression for the expected change in H within the simplex.

Where $i, j, k, N-i-j-k$ are the players playing R, P, S, and the 4th strategy respectively.

$\Delta H = H(t+1) - H(t)$,
$\Delta H = -x_{t+1}(1-x_{t+1}) - (-x_t(1-x_t))$
$\Delta H = -x_{t+1}(1-x_{t+1}) + x_t(1-x_t)$
$\Delta H = x_t(1-x_t) - x_{t+1}(1-x_{t+1})$

$\langle \Delta H \rangle = \sum_{i,j,k} \Big(H_s - H_{s'}\Big) T^{s \to s'}$, s is a particular state in the simplex.

$$\begin{aligned}
\langle \Delta H_{SD} \rangle = \frac{6}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& (N-i-j-k)(1-N+i+j+k)(T^{R+} + T^{P+} + T^{S+} + T^{+R} + T^{+P} + T^{+S}) \\
& - (N-i-j-k+1)(-N+i+j+k)T^{R+} \\
& - (N-i-j-k+1)(-N+i+j+k)T^{P+} \\
& - (N-i-j-k-1)(2-N+i+j+k)T^{+R} \\
& - (N-i-j-k-1)(2-N+i+j+k)T^{+P} \\
& - (N-i-j-k-1)(2-N+i+j+k)T^{+S} \Big]
\end{aligned} \tag{1}$$

The terms with transitions within the RPS simplex can be ignored as q would not change between these states, therefore the term $H_s - H_{s'} = 0$

$p = N-i-j-k$, the number of players playing the 4th strategy.

$$\begin{aligned}
\langle \Delta H_{SD} \rangle = \frac{6}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& p(1-p)(T^{R+} + T^{P+} + T^{S+} + T^{+R} + T^{+P} + T^{+S}) \\
& - (p+1)(-p)(T^{R+} + T^{P+} + T^{S+}) \\
& - (p-1)(2-p)(T^{+R} + T^{+P} + T^{+S}) \Big]
\end{aligned}$$

$$\begin{aligned}
\langle \Delta H_{SD} \rangle &= \frac{6}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[T^{R+} \left(p(1-p) + p(p+1)\right) + T^{+R} \left(p(1-p) - (p-1)(2-p)\right) \\
&\quad + T^{P+} \left(p(1-p) + p(p+1)\right) + T^{+P} \left(p(1-p) - (p-1)(2-p)\right) \\
&\quad + T^{S+} \left(p(1-p) + p(p+1)\right) + T^{+S} \left(p(1-p) - (p-1)(2-p)\right) \Big] \\
&= \frac{6}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[2p \left(T^{R+} + T^{P+} + T^{S+}\right) + (p - p^2 - (3p - p^2 - 2))(T^{+R} + T^{+P} + T^{+S}) \Big] \\
&= \frac{6}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[2p \left(T^{R+} + T^{P+} + T^{S+}\right) + (2-2p)(T^{+R} + T^{+P} + T^{+S}) \Big]
\end{aligned}$$

$$\langle \Delta H_{SD} \rangle = \frac{12}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[p \left(T^{R+} + T^{P+} + T^{S+} \right) + (1-p)(T^{+R} + T^{+P} + T^{+S}) \Big]$$

$$\langle \Delta H_{SD} \rangle = \frac{12}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} p \Big[(T^{R+} - T^{+R}) + (T^{P+} - T^{+P}) + (T^{S+} - T^{+S}) \Big] + T^{+R} + T^{+P} + T^{+S} \tag{2}$$

The continuous limit, where $x = i/N$, $y = j/N$, $z = k/N$, and $q = p/N$, $p = Nq$ and $q = 1 - x - y - z$ leads to:

$$\begin{aligned} \langle \Delta H_{SD} \rangle = \frac{12}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[& Nq \left[(T^{R+} - T^{+R}) + (T^{P+} - T^{+P}) + (T^{S+} - T^{+S}) \right] \\ & + (T^{+R} + T^{+P} + T^{+S}) \Big] \end{aligned}$$

Finally,

$$\begin{aligned} \langle \Delta H_{SD} \rangle = & \frac{12}{N} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[q \left[(T^{R+} - T^{+R}) + (T^{P+} - T^{+P}) + (T^{S+} - T^{+S}) \right] \Big] \\ & + \frac{12}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \left[T^{+R} + T^{+P} + T^{+S} \right] \end{aligned} \tag{3}$$

This can then be solved numerically and the critical population values can be found where $\langle \Delta H_{SD} \rangle = 0$.

Critical N found matches nicely with the simulated versions. Numerical integration in python code ./augRps.py, shows change of sign as expected. Matches nicely with the approximated values for the Moran process. The specific expression for Moran process is very long. Computed numerically and solved with scipy.integrate (reference scipy) Maybe can plot the simulated critical population sizes against the analyical on the same graph.

0.0.2 Analytical solution for the Local update process

Considering the local update process, an exact analytical solution can be found for $\langle \Delta H_{SD} \rangle$.

$$\begin{bmatrix} 0 & c & 1 & \gamma \\ 1 & 0 & c & \gamma \\ c & 1 & 0 & \gamma \\ \beta & \beta & \beta & 0 \end{bmatrix} \tag{4}$$

To simplify these derivations, I have fixed the values of $a = 0, b = 1$, allowing still to adjust the dynamics of the RPS game via c and the SD game via γ, β as used in all the results.

Evaluating the terms of the type $T^{R+} - T^{+R}$ for the local update process:

$$\begin{aligned} T^{R+} - T^{+R} &= \frac{i}{N} \frac{N-i-j-k}{N} \left(\frac{1}{2} + \frac{w}{2} \frac{\pi_+ - \pi_R}{\Delta \pi_{\max}} \right) - \frac{i}{N} \frac{N-i-j-k}{N} \left(\frac{1}{2} + \frac{w}{2} \frac{\pi_R - \pi_+}{\Delta \pi_{\max}} \right) \\ &= \frac{1}{2} \frac{i}{N} \frac{N-i-j-k}{N} w \left(\frac{\pi_+ - \pi_R}{\Delta \pi_{\max}} - \frac{\pi_R - \pi_+}{\Delta \pi_{\max}} \right) \\ &= w \frac{i}{N} \frac{N-i-j-k}{N} \left(\frac{\pi_+ - \pi_R}{\Delta \pi_{\max}} \right) \end{aligned}$$

substituting the population frequencies:

$$T^{R+} - T^{+R} = wx(1-x-y-z) \frac{\pi_+ - \pi_R}{\Delta \pi_{\max}} \tag{5}$$

the difference for the other two strategies can simply be derived by swapping the corresponding payoffs in the above expression.

Now considering the second integral with terms $T^{+R} + T^{+P} + T^{+S}$:

$$\begin{aligned} T^{+R} + T^{+P} + T^{+S} = & \frac{i}{N}\frac{N-i-j-k}{N}\left(\frac{1}{2} + \frac{w}{2}\frac{\pi_R - \pi_+}{\Delta\pi_{\max}}\right) \\ & + \frac{j}{N}\frac{N-i-j-k}{N}\left(\frac{1}{2} + \frac{w}{2}\frac{\pi_P - \pi_+}{\Delta\pi_{\max}}\right) \\ & + \frac{k}{N}\frac{N-i-j-k}{N}\left(\frac{1}{2} + \frac{w}{2}\frac{\pi_S - \pi_+}{\Delta\pi_{\max}}\right) \end{aligned}$$

Again replacing with the frequencies:

$$\begin{aligned} T^{+R} + T^{+P} + T^{+S} = & (1-x-y-z)\Bigg[x\left(\frac{1}{2} + \frac{w}{2}\frac{\pi_R - \pi_+}{\Delta\pi_{\max}}\right) \\ & + y\left(\frac{1}{2} + \frac{w}{2}\frac{\pi_P - \pi_+}{\Delta\pi_{\max}}\right) + z\left(\frac{1}{2} + \frac{w}{2}\frac{\pi_S - \pi_+}{\Delta\pi_{\max}}\right)\Bigg] \\ = & \frac{1}{2}(1-x-y-z)\left[x + y + z + \frac{w}{\Delta\pi_{\max}}\Big[x(\pi_R - \pi_+) + y(\pi_P - \pi_+) + z(\pi_S - \pi_+)\Big]\right] \end{aligned}$$

Substituting in the expanded expressions for payoff differences e.g. $\pi_R - \pi_+$:

$$\begin{aligned} T^{+R} + T^{+P} + T^{+S} = & \frac{1}{2}(1-x-y-z)\Bigg[x + y + z + \frac{w}{\Delta\pi_{\max}} \\ & \Big[x(z + cy + \gamma(1-x-yz) - \beta(x+y+z)) \\ & + y(x + cz + \gamma(1-x-yz) - \beta(x+y+z)) \\ & + z(y + cx + \gamma(1-x-yz) - \beta(x+y+z))\Big]\Bigg] \end{aligned}$$

Finally substituting everything back into $\langle \Delta H_{SD} \rangle$:

$$
\begin{aligned}
\langle \Delta H_{SD} \rangle = \frac{12}{N} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz (1-x-y-z) \Bigg[& \\
& \frac{w}{\Delta\pi_{\max}} x(1-x-y-z)\left(\beta(x+y+z) - \gamma(1-x-y-z) - z - cy\right) \\
& \frac{w}{\Delta\pi_{\max}} y(1-x-y-z)\left(\beta(x+y+z) - \gamma(1-x-y-z) - x - cz\right) \\
& \frac{w}{\Delta\pi_{\max}} z(1-x-y-z)\left(\beta(x+y+z) - \gamma(1-x-y-z) - y - cx\right) \Bigg] \\
& + \frac{6}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz (1-x-y-z) \Bigg[\\
& x+y+z+\frac{w}{\Delta\pi_{\max}} \Bigg(x(z+cy+\gamma(1-x-yz) - \beta(x+y+z)) \\
& + y(x+cz+\gamma(1-x-yz) - \beta(x+y+z)) \\
& + z(y+cx+\gamma(1-x-yz) - \beta(x+y+z)) \Bigg) \Bigg]
\end{aligned}
\tag{6}
$$

Solving the integrals leads to the exact analytical expression for local update process $\langle \Delta H_{SD} \rangle$:

$$
\langle \Delta H_{SD} \rangle_{LU} = \frac{42\Delta\pi_{\max} + 2w\gamma(7-6N) + (-1+4\beta-c)(-7+4N)w}{280N^2\Delta\pi_{\max}} \tag{7}
$$

which is nicely expressed by the actual values in the payoff matrix for the game, here I have fixed the values of $a = 0, b = 1$ so that the expected drift can be written as an expression of c, β, γ, N and w. $\Delta\pi_{\max}$ is the maximum payoff difference which will depend on the choice of parameters, however it is a constant and usually defined as $b - c$ or $1 - c$ when b is fixed to 1 in this case as mostly we consider $0 < \gamma, \beta < 1$ so the maximal difference is between the winning and losing case of the rock paper scissors sub game.

Since the critical population size is defined as a change in sign in $\langle \Delta H_{SD} \rangle$, the critical population size, N_c can also be derived by setting the above equation to 0 and solvign for N.

This leads to:

$$
\begin{aligned}
N_c &= \frac{-42\Delta\pi_{\max} + w(28\beta - 7c - 14\gamma - 7)}{w(16\beta - 4c - 12\gamma - 4)} \\
&= \frac{-42\Delta\pi_{\max}}{w(16\beta - 4c - 12\gamma - 4)} + \frac{28\beta - 7c - 14\gamma - 7}{16\beta - 4c - 12\gamma - 4}
\end{aligned}
$$

I have compared this value to the numerically solved solutions of the triple integral and the results match perfectly. For both the values of the integral and also the critical population size which was previously calculated using brentq root finding method. Due to increasing complexity in the Moran process, I have used numerical methods to compute the value of the integral and critical population size instead of a closed form solution.

See if its possible to also derive closed from solution for RPS and 4 strategy observation variables but this was a very long process for SD case so may be worth solving with numerical method. Definitely not for the Moran process.

0.0.3 $\langle \Delta H_{RPS} \rangle$ Derivation

$\langle \Delta H \rangle$ within the RPS plane $H = -xyz$.

$$
\begin{aligned}
\langle \Delta H_{RPS} \rangle = \frac{2}{N^6} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& ijk(T^{RP} + T^{RS} + T^{R+} + T^{PR} + T^{PS} + T^{P+} \\
& + T^{SR} + T^{SP} + T^{S+} + T^{+R} + T^{+P} + T^{+S}) \\
& - k(i-1)(j+1)T^{RP} - (i-1)j(k+1)T^{RS} - jk(i-1)T^{R+} \\
& - k(i+1)(j-1)T^{PR} - i(j-1)(k+1)T^{PS} - ik(j-1)T^{P+} \\
& - (i+1)j(k-1)T^{SR} - i(j+1)(k-1)T^{SP} - ij(k-1)T^{S+} \\
& - jk(i+1)T^{+R} - ik(j+1)T^{+P} - ij(k+1)T^{+S} \Big]
\end{aligned}
$$

$$
\begin{aligned}
\langle \Delta H_{RPS} \rangle = \frac{2}{N^6} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& k(j-i)(T^{RP} - T^{PR}) + j(k-i)(T^{RS} - T^{SR}) + i(j-k)(T^{SP} - T^{PS}) \\
& + k(T^{RP} + T^{PR}) + j(T^{RS} + T^{SR}) + i(T^{SP} + T^{PS}) \\
& + jk(T^{R+} - T^{+R}) + ik(T^{P+} - T^{+P}) + ij(T^{S+} - T^{+S}) \Big]
\end{aligned}
$$

$$
\begin{aligned}
\langle \Delta H_{RPS} \rangle = \frac{2}{N^3} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[& N^2 z(y-x)(T^{RP} - T^{PR}) + N^2 y(z-x)(T^{RS} - T^{SR}) \\
& + N^2 x(y-z)(T^{SP} - T^{PS}) \\
& + Nz(T^{RP} + T^{PR}) + Ny(T^{RS} + T^{SR}) + Nx(T^{SP} + T^{PS}) \\
& + N^2 xz(T^{P+} - T^{+P}) + N^2 yz(T^{R+} - T^{+R}) + N^2 xy(T^{S+} - T^{+S}) \Big]
\end{aligned}
$$

$$
\begin{aligned}
\langle \Delta H_{RPS} \rangle = \frac{2}{N} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[& z(y-x)(T^{RP} - T^{PR}) + y(z-x)(T^{RS} - T^{SR}) \\
& + x(y-z)(T^{SP} - T^{PS}) + xz(T^{P+} - T^{+P}) + yz(T^{R+} - T^{+R}) + xy(T^{S+} - T^{+S}) \Big] \\
+ \frac{2}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[& z(T^{RP} + T^{PR}) + y(T^{RS} + T^{SR}) + x(T^{SP} + T^{PS}) \Big]
\end{aligned}
$$

0.0.4 Analytical solution for the Local update process $\langle \Delta H_{RPS} \rangle$

An analytical solution for $\langle \Delta H_{RPS} \rangle$ can also be derived for for the local update process, this will require the simplified terms of the forms $T^{RP} - T^{PR}$ and $T^{RP} + T^{PR}$ within the RPS game, and the terms of form $T^{R+} - T^{+R}$ have already been derived in the solution for $\langle \Delta H_{SD} \rangle$.

$$
\begin{aligned}
T^{RP} - T^{PR} &= \left(\frac{1}{2} + \frac{w}{2} \left(\frac{\pi_P - \pi_R}{\Delta \pi_{max}} \right) \right) \frac{i}{N} \frac{j}{N} - \left(\frac{1}{2} + \frac{w}{2} \left(\frac{\pi_R - \pi_P}{\Delta \pi_{max}} \right) \right) \frac{i}{N} \frac{j}{N} \\
&= \frac{i}{N} \frac{j}{N} \left[\frac{w}{2\Delta \pi_{max}} \left(2(\pi_P - \pi_R) \right) \right] \\
&= \frac{w}{\Delta \pi_{max}} xy(\pi_P - \pi_R)
\end{aligned}
$$

where population counts e.g. $i/N, j/N$ are replaced with frequencies x, y respectively. Similarly the other 2 terms within the RPS game can be derived leading to:

$$
\begin{aligned}
T^{RS} - T^{SR} &= \frac{w}{\Delta \pi_{max}} xz(\pi_S - \pi_R) \\
T^{SP} - T^{PS} &= \frac{w}{\Delta \pi_{max}} zy(\pi_P - \pi_S)
\end{aligned}
$$

Substituting in the expanded payoff differences from the payoff matrix:

$$
\begin{aligned}
T^{RP} - T^{PR} &= \frac{w}{\Delta\pi_{max}} xy(x - z + c(z - y)) \\
T^{RS} - T^{SR} &= \frac{w}{\Delta\pi_{max}} xz(y - z + c(x - y)) \\
T^{SP} - T^{PS} &= \frac{w}{\Delta\pi_{max}} zy(x - y + c(z - x))
\end{aligned}
$$

The terms of form $T^{R+} - T^{+R}$ have already been derived in section for $\langle \Delta H_{SD} \rangle$.
Finally for the terms of order $1/N^2$:

$$
\begin{aligned}
T^{RP} + T^{PR} &= \left(\frac{1}{2} + \frac{w}{2}\left(\frac{\pi_P - \pi_R}{\Delta\pi_{max}}\right)\right)\frac{i}{N}\frac{j}{N} + \left(\frac{1}{2} + \frac{w}{2}\left(\frac{\pi_R - \pi_P}{\Delta\pi_{max}}\right)\right)\frac{i}{N}\frac{j}{N} \\
&= \frac{w}{2\Delta\pi_{max}} xy\left(\pi_P - \pi_R + \pi_R - \pi_p\right) + xy \\
&= xy
\end{aligned}
$$

The terms containing w and payoffs for each strategy cancel out for these terms within RPS, leaving just:

$$
\begin{aligned}
T^{RP} + T^{PR} &= xy \\
T^{RS} + T^{SR} &= xz \\
T^{RS} + T^{SR} &= zy
\end{aligned}
$$

$$
\begin{aligned}
\langle \Delta H_{RPS} \rangle_{LU} = \frac{2}{N}\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[& z(y-x)(\frac{w}{\Delta\pi_{max}} xy(x - z + c(z - y))) \\
& + y(z-x)(\frac{w}{\Delta\pi_{max}} xz(y - z + c(x - y))) + x(y-z)(\frac{w}{\Delta\pi_{max}} zy(x - y + c(z - x))) \\
& + xz(\frac{w}{\Delta\pi_{\max}} y(1-x-y-z)\left(\beta(x+y+z) - \gamma(1-x-y-z) - x - cz\right)) \\
& + yz(\frac{w}{\Delta\pi_{\max}} x(1-x-y-z)\left(\beta(x+y+z) - \gamma(1-x-y-z) - z - cy\right)) \\
& + xy(\frac{w}{\Delta\pi_{\max}} z(1-x-y-z)\left(\beta(x+y+z) - \gamma(1-x-y-z) - y - cx\right))\Big] \\
& + \frac{2}{N^2}\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz\,(3xyz)
\end{aligned}
$$

The second integral equals:

$$
\begin{aligned}
\frac{2}{N^2}\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz\,(3xyz) &= \frac{6}{N^2}\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz\,(xyz) \\
&= \frac{6}{N^2}\frac{1}{720} = \frac{1}{120N^2}
\end{aligned}
$$

$$
\begin{aligned}
\langle \Delta H_{RPS} \rangle_{LU} = & \frac{w}{\Delta\pi_{max}} \frac{2}{N} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz xyz \Big[(y-x)(x-z+c(z-y)) \\
& + (z-x)(y-z+c(x-y)) + (y-z)(x-y+c(z-x)) \\
& + (1-x-y-z)\left(\beta(x+y+z) - \gamma(1-x-y-z) - x - cz\right) \\
& + (1-x-y-z)\left(\beta(x+y+z) - \gamma(1-x-y-z) - z - cy\right) \\
& + (1-x-y-z)\left(\beta(x+y+z) - \gamma(1-x-y-z) - y - cx\right) \Big] \\
& + \frac{1}{120N^2} \\
= & \frac{1}{120N^2} - \frac{w(2+\gamma-3\beta+2c)}{3360N\Delta\pi_{max}}
\end{aligned} \tag{8}
$$

By setting this expression equal to zero, a critical population size for RPS can be found:

$$
N_{RPS} = \frac{28\Delta\pi_{max}}{w(2+\gamma-3\beta+2c)} \tag{9}
$$

0.0.5 Weak selection limit for the local update process for H_{SD}, H_{RPS}

Considering the weak selection limit where $w \to 0$, terms of the form $T^{R+} - T^{+R}$ vanish, so for the SD drift measure only the terms under $1/N^2$ are left.

The sum $T^{+R} + T^{+P} + T^{+S}$ simplifies to:

$$
\begin{aligned}
T^{+R} + T^{+P} + T^{+S} &= \frac{1}{2} \frac{N-i-j-k}{N} \left(\frac{i}{N} + \frac{j}{N} + \frac{k}{N} \right) \\
&= \frac{1}{2} q(x+y+z) = \frac{1}{2}(1-x-y-z)(x+y+z)
\end{aligned}
$$

Therefore $\langle \Delta H_{\mathrm{SD}} \rangle$ becomes (when $w \to 0$):

$$
\begin{aligned}
\langle \Delta H_{\mathrm{SD}} \rangle_{w\to 0} = & \frac{12}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \\
& \left[\frac{1}{2}(1-x-y-z)(x+y+z) \right] \\
= & \frac{3}{20N^2}
\end{aligned} \tag{10}
$$

As $w \to 0$ the three values of $\langle \Delta H_{SD} \rangle$ as population size varies will approach this limit and a drift reversal will no longer occur as the critical population size is undefined since $\frac{3}{20N^2} \neq 0$ at any point since $N \in \mathbb{N}$. However in the linear in w case even for weak selection where $w << 1/2$ a drift reversal is possible for each of the processes.

For $\langle \Delta H_{RPS} \rangle$, the terms of the form $T^{RP} - T^{PR}$ vanish, and the integral scaled by $2/N^2$ reduces to:

$$
\begin{aligned}
\frac{2}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \, (3xyz) & \\
= & \frac{1}{120N^2}
\end{aligned} \tag{11}
$$

like the SD observable, in the weak selection limit there is also no drift reversal as $\langle \Delta H_{RPS} \rangle$ does not change sign, however for $w > 0$, it is possible depending on the parameters of the system.

In the weak selection limit the difference in payoff between the strategies has little effect on reproduction relative to the random drift. This corresponds to the theory of neutral evolution which is popular in molecular evolution and states that the majority of evolution at the molecular level is governed by random drift as opposed to the differences in payoffs, or fitness of the individuals in the population.

0.0.6 $\langle \Delta H_4 \rangle$ Derivation

All 4 strategies, $H = -xyz(1-x-y-z)$, Therefore $\langle \Delta H_4 \rangle$ for the population counts can be defined as:

$$\begin{aligned}\langle \Delta H_4 \rangle = \frac{6}{N^7} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& ijk(N-i-j-k)(T^{RP} + T^{RS} + T^{R+} + T^{PR} + T^{PS} + T^{P+} \\ & + T^{SR} + T^{SP} + T^{S+} + T^{+R} + T^{+P} + T^{+S}) - (i-1)(j+1)k(N-i-j-k)T^{RP} \\ & - (i-1)j(k+1)(N-i-j-k)T^{RS} - (i-1)jk(N-i-j-k+1)T^{R+} \\ & - (i+1)(j-1)k(N-i-j-k)T^{PR} - i(j-1)(k+1)(N-i-j-k)T^{PS} \\ & - i(j-1)k(N-i-j-k+1)T^{P+} - (i+1)j(k-1)(N-i-j-k)T^{SR} \\ & - i(j+1)(k-1)(N-i-j-k)T^{SP} - ij(k-1)(N-i-j-k+1)T^{S+} \\ & - (i+1)jk(N-1-i-j-k)T^{+R} - i(j+1)k(N-1-i-j-k)T^{+P} - ij(k+1)(N-1-i-j-k)T^{+S} \Big]\end{aligned} \tag{12}$$

Normalization $\frac{6}{N^7}$, as its over the whole simplex (pyramid volume 1/6) and triple summation, (N^3), then 4 populations N^4.

Let p = N - i - j - k (4th strategy count):

$$\begin{aligned}\langle \Delta H_4 \rangle = \frac{6}{N^7} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& ijkp(T^{RP} + T^{RS} + T^{R+} + T^{PR} + T^{PS} + T^{P+} \\ & + T^{SR} + T^{SP} + T^{S+} + T^{+R} + T^{+P} + T^{+S}) - (i-1)(j+1)kpT^{RP} \\ & - (i-1)j(k+1)pT^{RS} - (i-1)jk(p+1)T^{R+} \\ & - (i+1)(j-1)kpT^{PR} - i(j-1)(k+1)pT^{PS} \\ & - i(j-1)k(p+1)T^{P+} - (i+1)j(k-1)pT^{SR} \\ & - i(j+1)(k-1)pT^{SP} - ij(k-1)(p+1)T^{S+} \\ & - (i+1)jk(p-1)T^{+R} - i(j+1)k(p-1)T^{+P} - ij(k+1)(p-1)T^{+S} \Big]\end{aligned} \tag{13}$$

Looking at one pair of transitions in and out of the same state. ($R \to P, P \to R$) this can be rewritten as before for $\langle \Delta H_{RPS} \rangle$ as differences of transition probabilities in and out of a state.

$$\begin{aligned}\langle \Delta H_4 \rangle = \frac{6}{N^7} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& kp(j-i)(T^{RP} - T^{PR}) + jp(k-i)(T^{RS} - T^{SR}) \\ & + ip(j-k)(T^{SP} - T^{PS}) + kp(T^{RP} + T^{PR}) + jp(T^{RS} + T^{SR}) \\ & + ip(T^{SP} + T^{PS}) + jk(p-i)(T^{R+} - T^{+R}) + ik(p-j)(T^{P+} - T^{+P}) \\ & + ij(p-k)(T^{S+} - T^{+S}) + jk(T^{R+} + T^{+R}) + ik(T^{P+} + T^{+P}) + ij(T^{S+} + T^{+S}) \Big]\end{aligned} \tag{14}$$

As before, converting this to the integral form where $x = i/N, y = j/N, z = k/N$ and $q = p/N, p = Nq$ where $q = 1 - x - y - z$ leads to:

$$
\begin{aligned}
\langle \Delta H_4 \rangle = \frac{6}{N^4} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[& N^3 zq(y-x)(T^{RP} - T^{PR}) + N^3 yq(z-x)(T^{RS} - T^{SR}) \\
& + N^3 xq(y-z)(T^{SP} - T^{PS}) + N^2 zq(T^{RP} + T^{PR}) + N^2 yq(T^{RS} + T^{SR}) + N^2 xq(T^{SP} + T^{PS}) \\
& + N^3 yz(q-x)(T^{R+} - T^{+R}) + N^3 xz(q-y)(T^{P+} - T^{+P}) + N^3 xy(q-z)(T^{S+} - T^{+S}) \\
& + N^2 yz(T^{R+} + T^{+R}) + N^2 xz(T^{P+} + T^{+P}) + N^2 xy(T^{S+} + T^{+S}) \Big]
\end{aligned} \tag{15}
$$

$$
\begin{aligned}
\langle \Delta H_4 \rangle = \frac{6}{N} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[& zq(y-x)(T^{RP} - T^{PR}) + yq(z-x)(T^{RS} - T^{SR}) + xq(y-z)(T^{SP} - T^{PS}) \\
& + yz(q-x)(T^{R+} - T^{+R}) + xz(q-y)(T^{P+} - T^{+P}) + xy(q-z)(T^{S+} - T^{+S}) \Big] \\
+ \frac{6}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[& zq(T^{RP} + T^{PR}) + yq(T^{RS} + T^{SR}) + xq(T^{SP} + T^{PS}) \\
& + yz(T^{R+} + T^{+R}) + xz(T^{P+} + T^{+P}) + xy(T^{S+} + T^{+S}) \Big]
\end{aligned} \tag{16}
$$

Final derivation of $\langle \Delta H_4 \rangle$. There are clear similarities to the other two expected values $\langle \Delta H_{SD} \rangle, \langle \Delta_{RPS} \rangle$ with some terms missing due to the different observation variable H used but they can all be rewritten in terms of differences of transition probabilities into and out of state (e.g. $(T^{RP} - T^{PR})$) in an integral scaled by $1/N$ plus positive terms in a second integral scaled by $1/N^2$.

0.1 Figures for drift metric

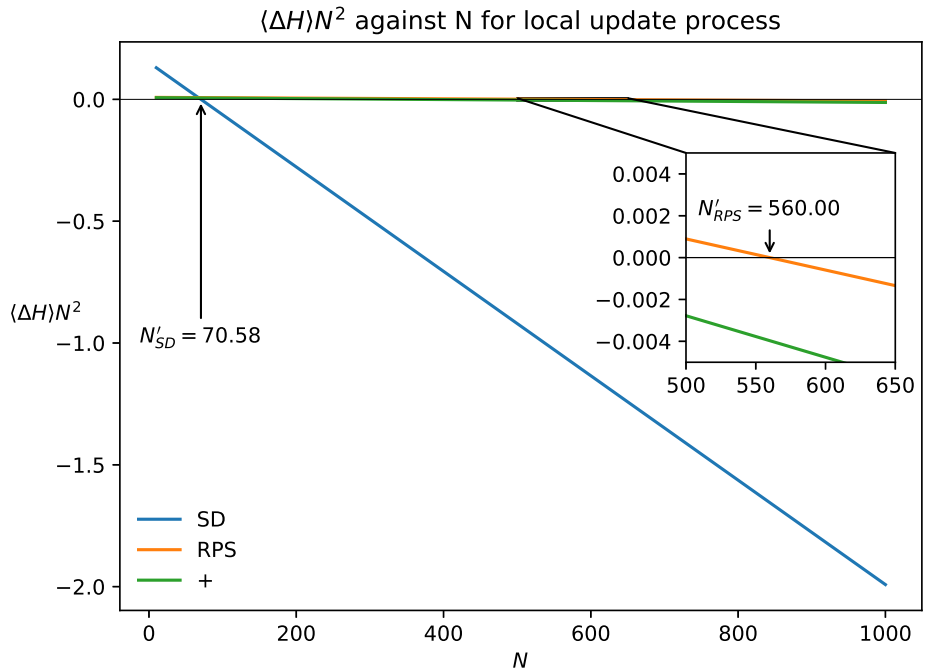


Figure 1: Comparison of drift measures for local update process, where $\beta = 0.4, \gamma = 0.2, c = -0.8$. Still an interior attracting RPS in large population case, however this time RPS drifts at a higher population size marked by N'_{RPS} so drifts to RPS boundaries whilst SD does not reverse and stays at the fixed point for SD part of the game. Results in disk shape around the central fixed point as RPS goes to boundary while SD is stable in the interior.

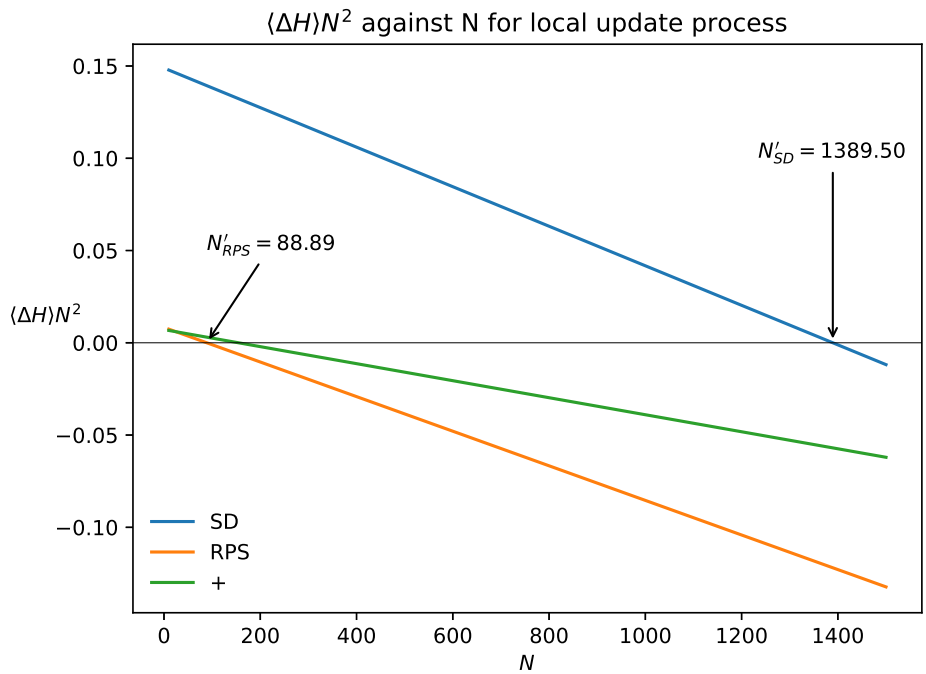


Figure 2: Comparison of the 3 drift measures $\langle \Delta H_{SD} \rangle, \langle \Delta H_{RPS} \rangle$, and$\langle \Delta H_{+} \rangle$ for the local update process. $\beta = 0.3, \gamma = 0.14, c = -0.2$, strong central attracting RPS case ($c > -1$) and $N_{RPS} < N_{SD}$ so SD game component reverses first, at a higher population size. This results in the vertical column point cloud where RPS fixates at the central fixed point where $x \approx y \approx z$ and q drifts away from the fixed point ($N_{RPS} < N < N_{SD}$)

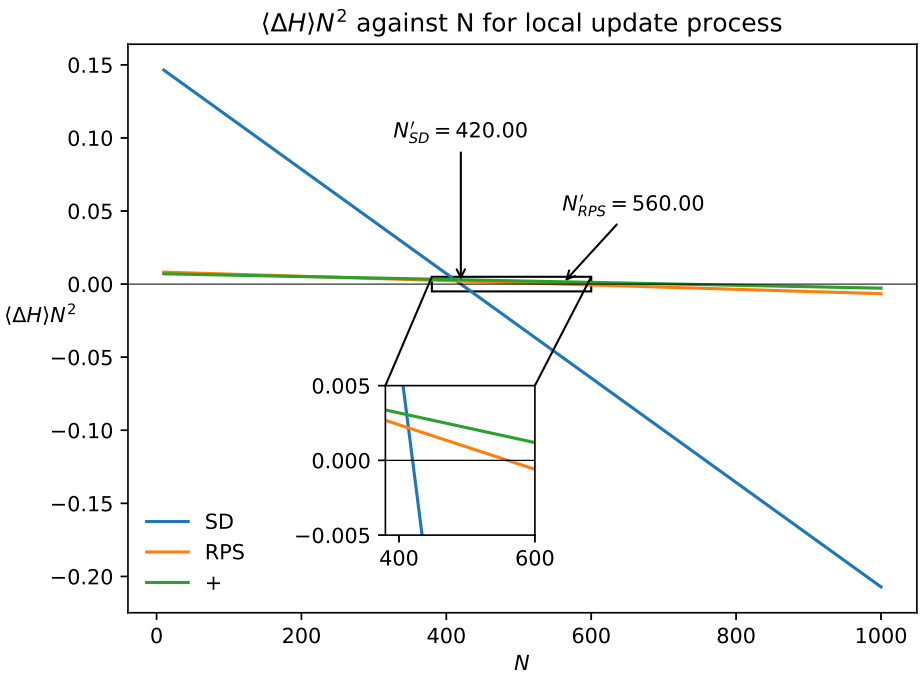


Figure 3: $\beta = 0.1, \gamma = 0.1, c = -0.8, N_{SD} = 420, N_{RPS} = 560$

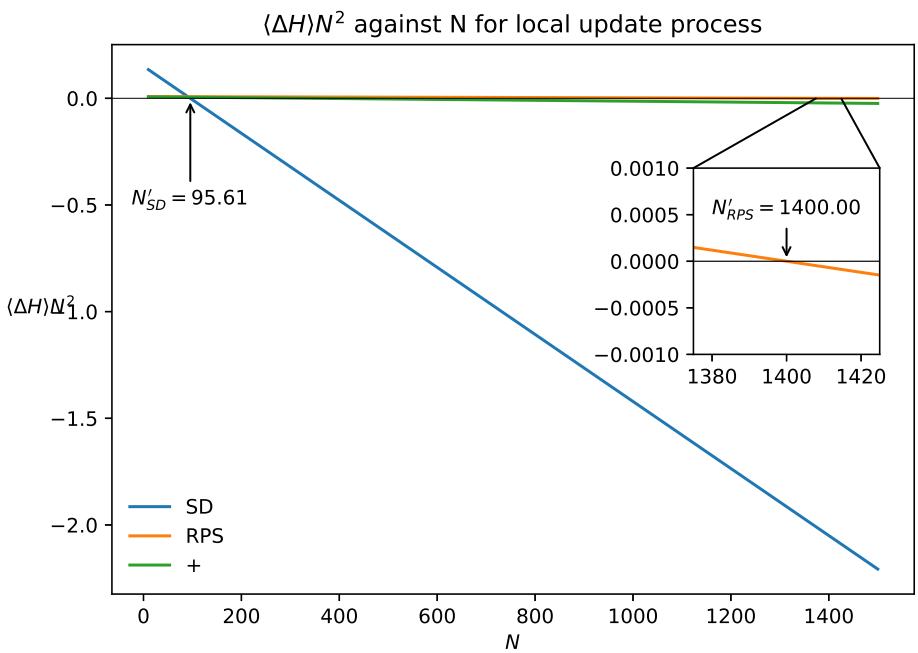


Figure 4: $\beta = 0.24, \gamma = 0.4, c = -0.8$ cleaner disk case, clamped the starting values where $q > 0.1 < 0.9$ to prevent RPS random extinctions for a clearer image. This parameter settings give a better plot becaues $N_{RPS} >> N_{SD}$ for the disk case. $N_{SD} = 95.61, N_{RPS} = 1400$

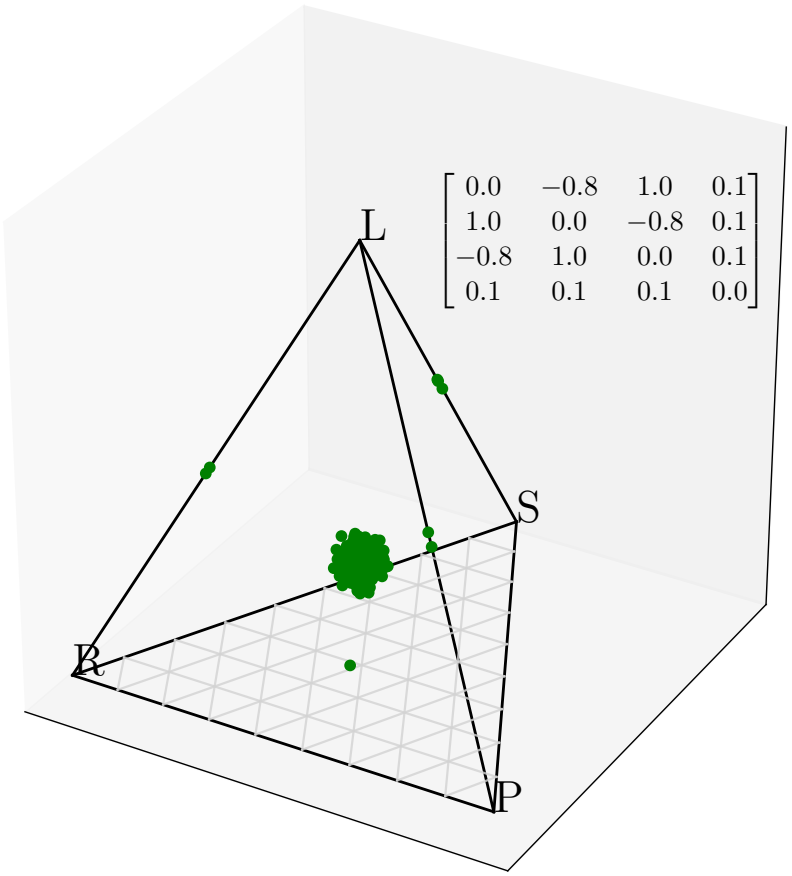


Figure 5: Large N 30000

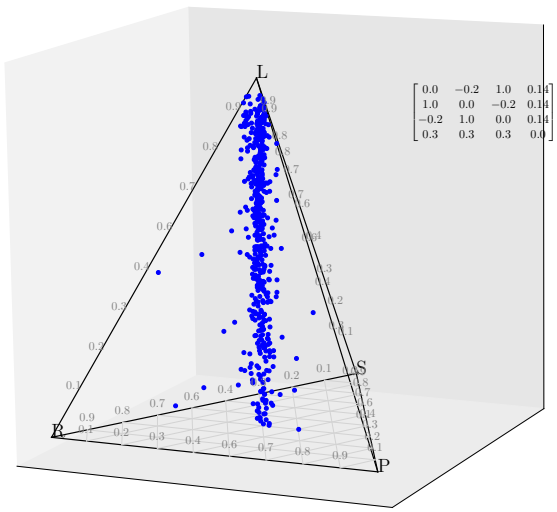


Figure 6: SD drift case N = 200

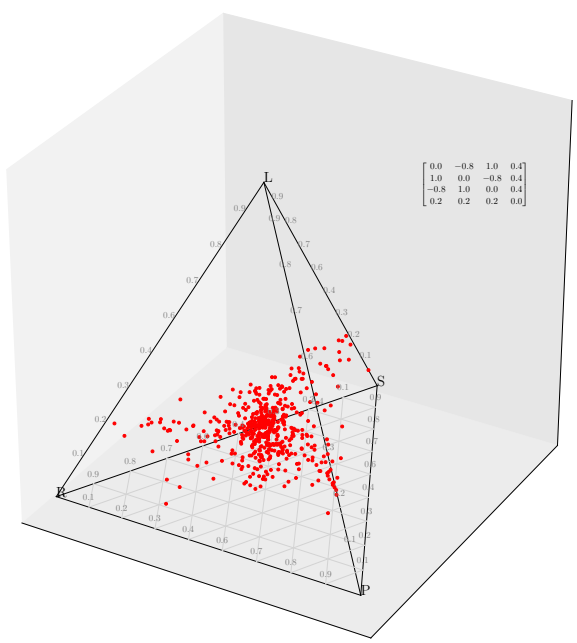


Figure 7: RPS drift case N = 300

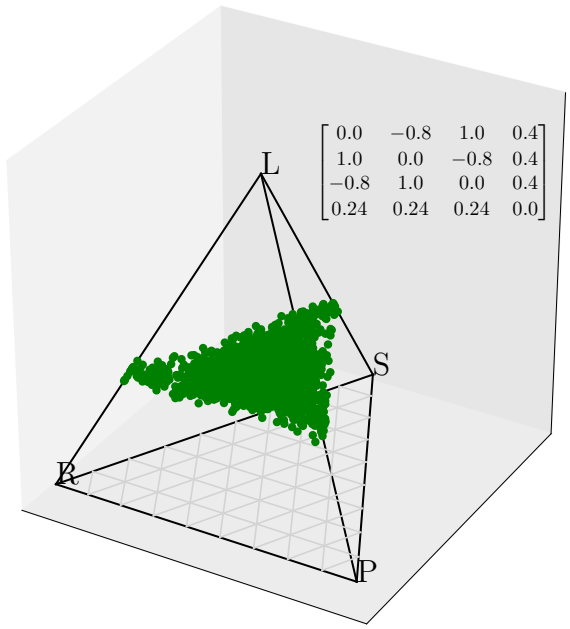


Figure 8: Cleaner drift case N = 600

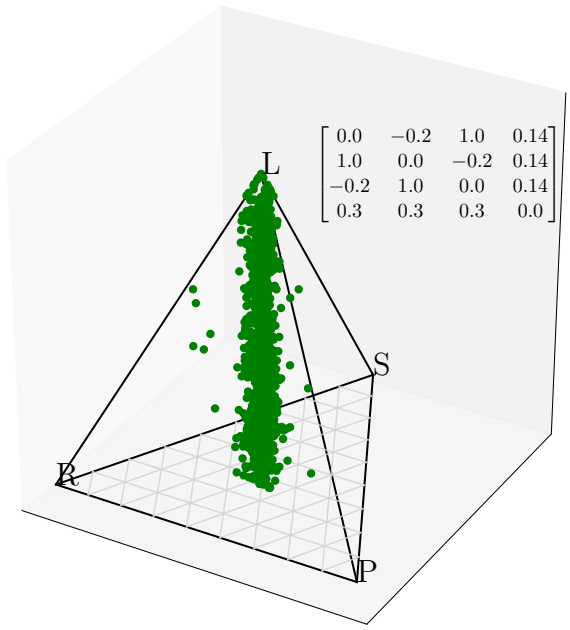


Figure 9: N = 300, rod

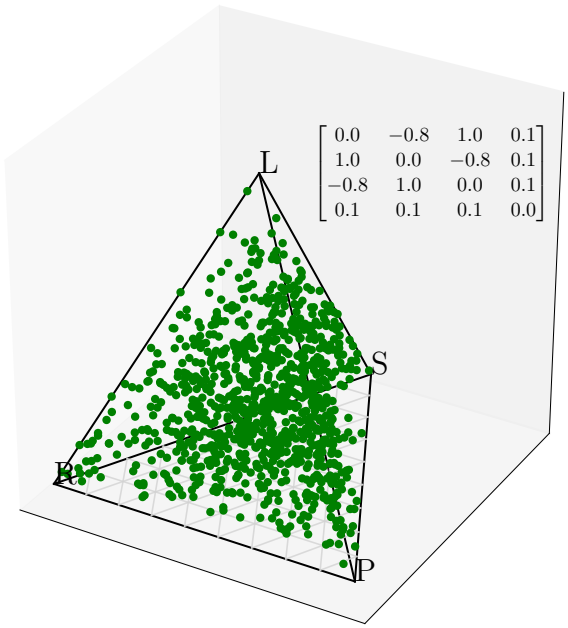


Figure 10: N = 50, same parameters as the Large N case

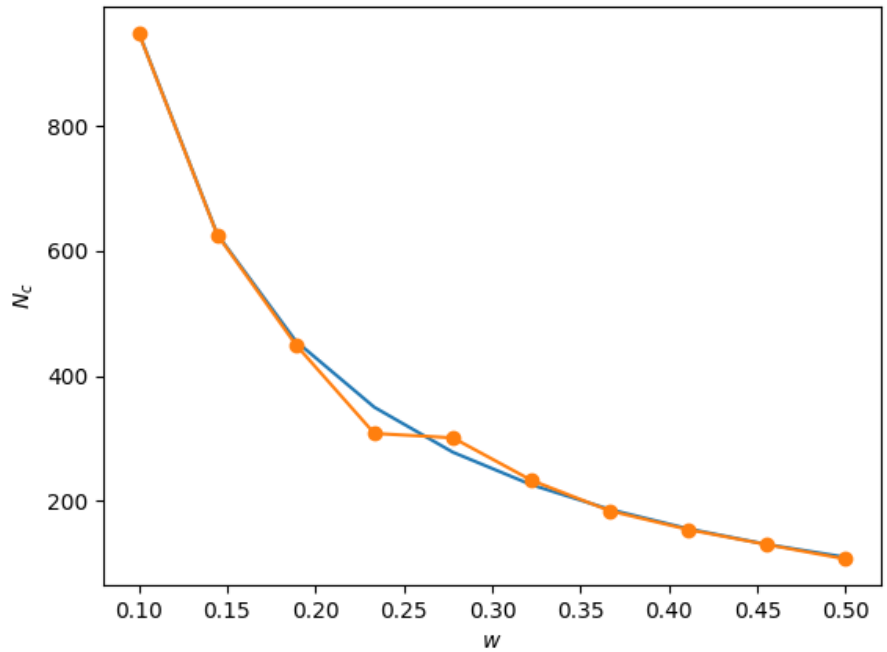


Figure 11: Comparison of simulated critical population sizes against the derived expected change in H for Moran process. Small deviations due to stochasticity in the simulations. $\gamma = 0.2, \beta = 0.1$ standard RPS. Small deviations due to random nature of simulations, but close match to analytical result. Drift reversal is present in the agent based simulations and exactly matching analytical prediction for these paremeters.

0.2 Drift reversal cases

Testing point cloud at N = 200 $\frac{N}{N_{SD}} = 0.14$, $\frac{N}{N_{RPS}} = 2.26$, rod case

For below at N = 300 $\frac{N}{N_{SD}} = 4.25$, $\frac{N}{N_{RPS}} = 0.54$

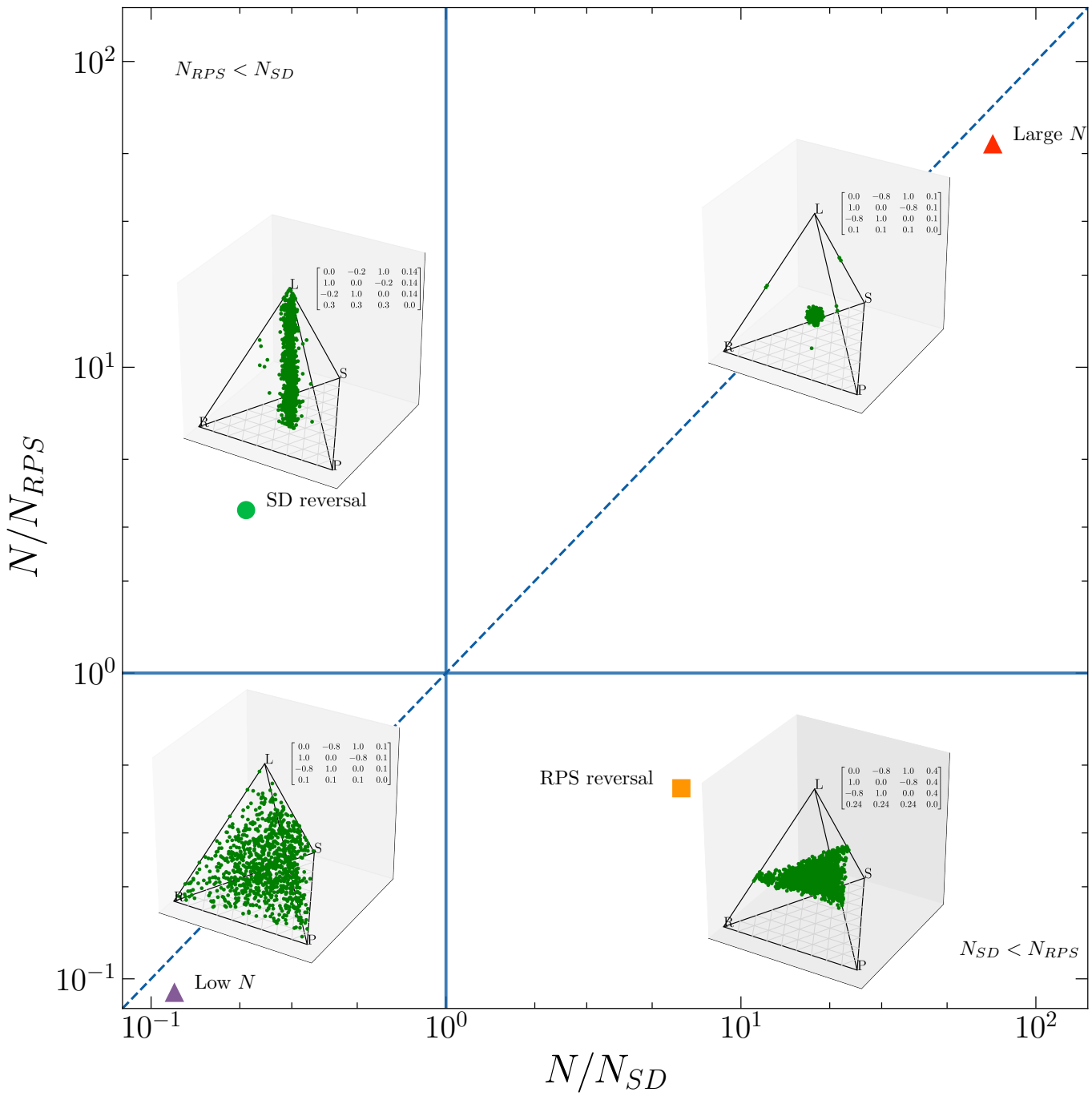
For final case at N = 2000 $\frac{N}{N_{SD}} = 4.76$, $\frac{N}{N_{RPS}} = 3.57$

For clean disk case N = 600, $\gamma = 0.4, \beta = 0.24, N_{SD} = 95.61, N_{RPS} = 1400$

Blob case N = 1000 $\frac{N}{N_{SD}} = 2.38$, $\frac{N}{N_{RPS}} = 1.79$ Final case at N = 30000, $\frac{N}{N_{SD}} = 71.43$, $\frac{N}{N_{RPS}} = 53.57$

α	β	c	N	N/N_{SD}	N/N_{RPS}
0.1	0.1	-0.8	50	0.12	0.09*
0.14	0.3	-0.2	200	0.14	2.26
0.14	0.3	-0.2	300	0.21	3.41*
0.4	0.2	-0.8	200	2.83	0.36
0.4	0.24	-0.8	600	6.28	0.42*
0.1	0.1	-0.8	2000	4.76	3.57
0.1	0.1	-0.8	30000	71.43	53.57*

Table 1: Table of N values and corresponding ratios, depending on the values of N_{SD}, N_{RPS} and N when simulating trajectories, the drift reversal cases will be different, large N used to show the cases above the critical population sizes where system converges to interior nash equilibrium. Rows marked with * are the ones in the drift cases plot.


$N_{RPS} < N_{SD}$
Large N
SD reversal
RPS reversal
Low N
$N_{SD} < N_{RPS}$
N/N_{RPS}
N/N_{SD}
10^{-1}
10^{0}
10^{1}
10^{2}
L
S
R
P